Matrices (AI HL)

ClipSAT Printable Practice Suite (clipsat.org) • Topic 6.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The temperature θ of a cooling metal rod satisfies $\frac{d\theta}{dt} = -k(\theta - 20)$, where t is time in minutes and k is a constant. Given $\theta = 80$ when $t = 0$ and $\theta = 50$ when $t = 10$, find the value of t when $\theta = 30$.

Separate variables and integrate: $\int \frac{1}{\theta - 20}\, d\theta = -\int k\, dt \Rightarrow \ln(\theta - 20) = -kt + C$ Apply initial condition $t = 0, \theta = 80$: $\ln(60) = C \Rightarrow \ln\left(\frac{\theta - 20}{60}\right) = -kt \Rightarrow \theta - 20 = 60e^{-kt}$ Apply condition $t = 10, \theta = 50$: $30 = 60e^{-10k} \Rightarrow e^{-10k} = \frac{1}{2} \Rightarrow k = \frac{\ln 2}{10} \approx 0.06931$ Set $\theta = 30$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the constant term (independent of x) in the binomial expansion of $\left(2x^2 - \frac{1}{x}\right)^9$.

Step 1: Write the general term T_{k+1}: $T_{k+1} = \binom{9}{k}(2x^2)^{9-k}\left(-\frac{1}{x}\right)^k = \binom{9}{k}2^{9-k}x^{2(9-k)}(-1)^k x^{-k} = \binom{9}{k}2^{9-k}(-1)^k x^{18-3k}.$ Step 2: Find k for the term independent of x: Set exponent equal to 0: $18 - 3k = 0 \Rightarrow 3k = 18 \Rightarrow k = 6.$ Step 3: Evaluate coefficient at $k = 6$: $\binom{9}{6} = \binom{9}{3} = \frac{9 \times 8 \times 7}{3 \times 2 \times 1} = 84.$.

3 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

The curve C has equation $x^3 - 2xy + y^2 = 4$. Find the coordinates of the two points on C where the tangent is parallel to the y-axis (vertical).

Differentiate implicitly with respect to x: $3x^2 - 2\left(y + x\frac{dy}{dx}\right) + 2y\frac{dy}{dx} = 0$ $3x^2 - 2y + (2y - 2x)\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = \frac{2y - 3x^2}{2y - 2x}$ A vertical tangent occurs when the denominator is zero (and numerator non-zero): $2y - 2x = 0 \Rightarrow y = x$ Substitute $y = x$ into the original curve equation: $x^3 - 2x(x) + x^2 = 4 \Rightarrow x^3 - x^2 - 4 = 0$ Test integer factors: $x = 2$ gives $8 - 4 - 4 = 0$. Factoring gives $(x - 2)(x^2 + x + 2) = 0$.

4 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A bag contains 5 red and 4 green marbles. Three marbles are drawn at random without replacement. Find the probability that: (a) all three are red, (b) at least two are red.

Total marbles = 9. Number of ways to choose 3 marbles = $\binom{9}{3} = \frac{9 \times 8 \times 7}{3 \times 2 \times 1} = 84$. (a) All 3 are red: $\binom{5}{3} = 10$. Probability: $P(\text{all red}) = \frac{10}{84} = \frac{5}{42} \approx 0.119$ (b) 'At least two red' means exactly 2 red and 1 green, or exactly 3 red: Ways for 2 red, 1 green: $\binom{5}{2} \times \binom{4}{1} = 10 \times 4 = 40$. Ways for 3 red: $\binom{5}{3} = 10$. Total favorable outcomes: $40 + 10 = 50$. Probability.

© ClipSAT — clipsat.org | All Rights Reserved

5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A train accelerates uniformly from rest to 30 m/s in 20 s, travels at constant speed 30 m/s for 60 s, and then decelerates uniformly to rest in 15 s. Find: (a) the acceleration in the first phase, (b) the total distance travelled, (c) the average speed for the entire journey.

(a) $a_1 = \frac{v-u}{t} = \frac{30-0}{20} = 1.5$ m/s^2. (b) Total distance is the area of the trapezoid on the v-t graph with parallel sides 60 s and $20+60+15 = 95$ s, and height 30 m/s: $s = \frac{60+95}{2} \times 30 = \frac{155}{2} \times 30 = 155 \times 15 = 2325$ m (c) Average speed.

6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A block of mass 4 kg is pulled along a rough horizontal plane ($\mu = 0.25$) by a pulling force of magnitude 25 N acting at 30° above the horizontal. Taking $g = 9.8$ m/s^2: (a) show that the normal reaction $R = 26.7$ N, (b) find the acceleration of the block.

(a) Resolve vertically (equilibrium in vertical direction): $R + 25\sin 30^\circ = mg \Rightarrow R + 25(0.5) = 4(9.8) \Rightarrow R + 12.5 = 39.2 \Rightarrow R = 26.7$ N (b) Limiting friction force opposes motion: $F = \mu R = 0.25 \times 26.7 = 6.675$ N Resolve horizontally using $F_{\text{net}} = ma$: $25\cos 30^\circ - F = ma \Rightarrow 25\left(\frac{\sqrt{3}}{2}\right) - 6.675 = 4a$.

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Two particles P of mass 3 kg and Q of mass 5 kg are connected by a light inextensible string passing over a smooth fixed pulley. The system is held at rest and released. Taking $g = 9.8$ m/s^2: (a) find the acceleration a of the system, (b) calculate the tension T in the string, (c) find the force exerted by the string on the pulley.

(a) Write $F = ma$ for each particle in its direction of motion: For Q (moving downwards): $5g - T = 5a$ For P (moving upwards): $T - 3g = 3a$ Add the equations: $2g = 8a \Rightarrow a = \frac{2(9.8)}{8} = 2.45$ m/s^2 (b) Substitute a into the equation for P: $T = 3g + 3a = 3(9.8 + 2.45) = 3(12.25) = 36.75$ N ≈ 36.8 N (c) The pulley supports two vertical segments of string with tension T, so the total downward force on the pulley is.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A continuous random variable X has PDF given by $f(x) = \frac{k}{x^2}$ for $1 \le x \le 4$, and 0 elsewhere. (a) Show that $k = \frac{4}{3}$. (b) Find the median of X.

(a) Integrate over the support and equate to 1: $\int_1^4 \frac{k}{x^2}\,dx = k\left[-\frac{1}{x}\right]_1^4 = k\left(-\frac{1}{4} - (-1)\right) = \frac{3}{4}k = 1 \Rightarrow k = \frac{4}{3}$ (b) Find the CDF for $1 \le x \le 4$: $F(x) = \int_1^x \frac{4}{3t^2}\,dt = \frac{4}{3}\left(1 - \frac{1}{x}\right)$ Set $F(m) = 0.5$ to find the median m.

© ClipSAT — clipsat.org | All Rights Reserved

9 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Write the standard equation of the ellipse having foci at $(\pm 3, 0)$ and vertices at $(\pm 5, 0)$. State the lengths of the major and minor axes, and compute the eccentricity.

Step 1: Identify ellipse parameters: The vertices and foci lie on the x-axis, so the major axis is horizontal and centered at $(0, 0)$. Here semi-major axis $a = 5$ and focal distance $c = 3$. Step 2: Compute semi-minor axis b: For an ellipse: $b^2 = a^2 - c^2 = 5^2 - 3^2 = 25 - 9 = 16 \Rightarrow b = 4$. Step 3: Formulate equation and invariants: $\frac{x^2}{25} + \frac{y^2}{16} = 1$. Major axis length $2a = 10$, minor axis length $2b = 8$, and eccentricity $e = \frac{c}{a} = \frac{3}{5} = 0.6$.

10 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

An ellipse has vertices at $(1, 6)$ and $(1, -4)$, and a focus at $(1, 4)$. Determine its standard equation and compute its eccentricity.

The vertices share $x = 1$, so the major axis is vertical. Center is the midpoint of the vertices: $(h, k) = \left(1, \frac{6 + (-4)}{2}\right) = (1, 1)$. Semi-major axis: $a = 6 - 1 = 5$ (so $a^2 = 25$). Focal distance from center to $(1, 4)$: $c = 4 - 1 = 3$. For an ellipse, $b^2 = a^2 - c^2 = 25 - 9 = 16 \Rightarrow b = 4$. Standard equation: $\frac{(x-1)^2}{16} + \frac{(y-1)^2}{25} = 1$. Eccentricity: $e = \frac{c}{a} = \frac{3}{5} = 0.6$.

11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

A car of mass 1200 kg ascends a hill of inclination α, where $\sin\alpha = \frac{1}{20}$, against a constant resistance to motion of 500 N. Taking $g = 9.8$ m/s^2: (a) find the driving force required to travel at a steady speed of 25 m/s, (b) calculate the power developed by the engine in kilowatts.

(a) Component of weight down the slope: $W_{\parallel} = mg\sin\alpha = 1200(9.8)\left(\frac{1}{20}\right) = 60(9.8) = 588$ N Since the car travels at steady speed (acceleration $a = 0$), the engine driving force balances all resistance: $F_{\text{drive}} = \text{Resistance} + mg\sin\alpha = 500 + 588 = 1088$ N (b) Power output: $P = F_{\text{drive}} \times v = 1088 \times 25 = 27\,200$ W $= 27.2$ kW

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

A radioactive substance follows the decay model $M(t) = M_0 e^{-kt}$. If 20% of the sample decays in 15 days, find the exact decay constant k and the substance's half-life in days to the nearest tenth.

If 20% decays in 15 days, then 80% remains at $t = 15$: $0.80M_0 = M_0 e^{-15k} \Rightarrow e^{-15k} = 0.80$. Take the natural logarithm of both sides: $-15k = \ln(0.80) \Rightarrow k = -\frac{\ln(0.80)}{15} = \frac{\ln(1.25)}{15} \approx 0.01488$ day^{-1}. The half-life $t_{1/2}$ satisfies $e^{-kt_{1/2}} = 0.5$: $t_{1/2} = \frac{\ln 2}{k} = \frac{15\ln 2}{-\ln(0.80)} \approx \frac{15(0.6931)}{0.2231} \approx 46.6$ days.

© ClipSAT — clipsat.org | All Rights Reserved

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

On an Argand diagram, the point P represents the complex number z such that $|z-(2+3i)| \leq 2$. Find the greatest possible value of $|z|$ for points in this region.

The locus $|z-(2+3i)| \leq 2$ represents a circular disc of radius $R=2$ with center at $C(2,3)$. The distance from the origin $O(0,0)$ to the center C is: $|C| = \sqrt{2^2+3^2} = \sqrt{4+9} = \sqrt{13} \approx 3.6055$ The point on the disc farthest from the origin lies on the line passing through O and C, on the far boundary of the circle: $|z|_{\max} = |C| + R = \sqrt{13} + 2 \approx 3.606 + 2 = 5.61$ (3 s. f.)

14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Events A and B are such that $P(A)=0.6$, $P(B)=0.5$, and $P(A \cup B)=0.8$. Find $P(A \cap B)$, calculate $P(A|B)$, and determine whether A and B are independent.

1. Use the addition rule: $P(A \cup B) = P(A) + P(B) - P(A \cap B) \Rightarrow 0.8 = 0.6 + 0.5 - P(A \cap B) \Rightarrow P(A \cap B) = 1.1 - 0.8 = 0.3$. 2. Conditional probability: $P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{0.3}{0.5} = 0.6$ 3. Check for independence: $P(A \cap B) = 0.3$ and $P(A) \times P(B) = 0.6 \times 0.5 = 0.30$. Since $P(A \cap B) = P(A)P(B)$ (or equivalently $P(A|B) = P(A) = 0.6$), events A and B are independent.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)** • **Cross-Lesson: 5.02 Differential Equations**

Find the equation of the perpendicular bisector of the line segment joining $A(-2,5)$ and $B(4,\ -3)$. Give your answer in the form $ax+by+c=0$, where a, b, c are integers.

1. Midpoint of AB: $M = \left(\frac{-2+4}{2}, \frac{5-3}{2}\right) = (1,1)$. 2. Gradient of AB: $m_{AB} = \frac{-3-5}{4-(-2)} = \frac{-8}{6} = -\frac{4}{3}$. 3. Gradient of perpendicular bisector: $m_{\perp} = -\frac{1}{-4/3} = \frac{3}{4}$. 4. Equation of line passing through $M(1,1)$: $y-1 = \frac{3}{4}(x-1) \Rightarrow 4(y-1) = 3(x-1) \Rightarrow 4y-4 = 3x-3$ Rearrange to standard integer form: $3x-4y+1=0$.

16 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)** • **Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series**

The random variable $X \sim N(\mu, \sigma^2)$ satisfies $P(X<65)=0.8413$ and $P(X<45)=0.0228$. Find the values of μ and σ.

Standardising: $\Phi\left(\frac{65-\mu}{\sigma}\right) = 0.8413 \Rightarrow \frac{65-\mu}{\sigma} = 1.0$, so $65-\mu=\sigma$. For the lower tail: $P(X<45) = 0.0228 \Rightarrow \Phi\left(\frac{45-\mu}{\sigma}\right) = 0.0228 \Rightarrow \frac{45-\mu}{\sigma} = -2.0$, so $\mu - 45 = 2\sigma$. Adding the two equations: $65-45 = 3\sigma \Rightarrow 20 = 3\sigma \Rightarrow \sigma = 6.67$. Then $\mu = 65 - 6.67 = 58.33$.

© ClipSAT — clipsat.org | All Rights Reserved

17 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)** • **Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)**

Express $\frac{3+\sqrt{5}}{2-\sqrt{5}}$ in the form $p+q\sqrt{5}$, where p and q are rational numbers to be found.

Multiply numerator and denominator by the conjugate of the denominator, $2+\sqrt{5}$: $\frac{3+\sqrt{5}}{2-\sqrt{5}} \times \frac{2+\sqrt{5}}{2+\sqrt{5}} = \frac{(3+\sqrt{5})(2+\sqrt{5})}{2^2-(\sqrt{5})^2}$ Expanding the numerator: $3(2)+3\sqrt{5}+2\sqrt{5}+(\sqrt{5})^2 = 6+5\sqrt{5}+5 = 11+5\sqrt{5}$. The denominator is $4-5=-1$. Therefore: $\frac{11+5\sqrt{5}}{-1} = -11-5\sqrt{5}$ Thus $p=-11$ and $q=-5$.

18 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)** • **Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)**

Level 2 · Medium Solve the initial value problem $\frac{dy}{dx} - y = e^{3x}$ with $y(0)=2$.

Identify $P(x)=-1$ and $Q(x)=e^{3x}$. Compute integrating factor: $\mu(x) = e^{\int -1\,dx} = e^{-x}$. Multiply through: $\frac{d}{dx}[e^{-x}y] = e^{-x}e^{3x} = e^{2x}$ Integrate both sides: $e^{-x}y = \int e^{2x}\,dx = \frac{1}{2}e^{2x}+C \Rightarrow y(x) = \frac{1}{2}e^{3x}+Ce^{x}$ Apply initial condition $y(0)=2$: $2=\frac{1}{2}(1)+C(1) \Rightarrow C=\frac{3}{2}$ Thus, $y(x)=\frac{1}{2}e^{3x}+\frac{3}{2}e^{x}$.

19 **[Challenging]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)** • **Cross-Lesson: 6.03 Markov Chains (AI HL)**

Level 1 · FoundationsExpress the infinite summation limit as a definite integral on the interval $[0,1]$ and evaluate it: $\lim_{n\to\infty} \sum_{i=1}^{n} \frac{i^3}{n^4}$.

Factor $\frac{1}{n}=\Delta x$: $\sum_{i=1}^{n} \frac{i^3}{n^4} = \sum_{i=1}^{n} \left(\frac{i}{n}\right)^3 \cdot \frac{1}{n}$. Here $x_i = \frac{i}{n} = 0 + i\Delta x$ with $\Delta x = \frac{1}{n}$, matching the right Riemann sum of $f(x)=x^3$ on $[0,1]$: $\lim_{n\to\infty} \sum_{i=1}^{n} \left(\frac{i}{n}\right)^3 \frac{1}{n} = \int_0^1 x^3\,dx = \left[\frac{x^4}{4}\right]_0^1 = \frac{1^4}{4} - 0 = \mathbf{\frac{1}{4}}$.

20 **[Challenging]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)** • **Cross-Lesson: 1.01 Proof by Mathematical Induction**

Solve the logarithmic equation $\log_2(x+3)+\log_2(x-3)=4$ for all real solutions x. State any extraneous roots.

Step 1: State domain restrictions: $x+3>0 \Rightarrow x>-3$ and $x-3>0 \Rightarrow x>3$, so $x>3$. Step 2: Combine logarithms: $\log_2[(x+3)(x-3)]=4 \Rightarrow \log_2(x^2-9)=4$. $x^2-9=2^4=16 \Rightarrow x^2=25 \Rightarrow x=\pm 5$. Step 3: Eliminate extraneous solutions: $x=-5$ violates $x>3$ (producing negative arguments), so it is extraneous. The unique valid solution is $x=5$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

© ClipSAT — clipsat.org | All Rights Reserved

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Compute the matrix product AB for $A = [[1, 2], [3, 4]]$ and $B = [[2, 0], [1, 3]]$.

a. $[[4, 6], [10, 12]]$ ✓

b. $[[2, 0], [3, 12]]$

c. $[[4, 10], [6, 12]]$

d. $[[3, 2], [4, 7]]$

Row 1: $[1(2) + 2(1), 1(0) + 2(3)] = [4, 6]$. Row 2: $[3(2) + 4(1), 3(0) + 4(3)] = [10, 12]$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the inverse of the matrix $M = [[3, 5], [1, 2]]$.

a. $[[2, -5], [-1, 3]]$ ✓

b. $[[-2, 5], [1, -3]]$

c. $[[3, -1], [-5, 2]]$

d. $[[1/3, 1/5], [1, 1/2]]$

$\det(M) = 3(2) - 5(1) = 6 - 5 = 1$. The inverse is $\frac{1}{\det}[[d, -b], [-c, a]] = [[2, -5], [-1, 3]]$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If A and B are invertible matrices of the same dimension, what is $(AB)^{-1}$?

a. $B^{-1}A^{-1}$ ✓

b. $A^{-1}B^{-1}$

c. BA^{-1}

d. AB^{-1}

The socks-and-shoes property of matrix inverses reverses order: $(AB)^{-1} = B^{-1}A^{-1}$.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For what value of k is the matrix $A = [[k, 6], [2, 3]]$ singular (non-invertible)?

a. $k = 4$ ✓

b. $k = 6$

c. $k = 0$

d. $k = 9$

A matrix is singular when its determinant is zero: $\det(A) = 3k - 12 = 0 \Rightarrow 3k = 12 \Rightarrow k = 4$.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If A is a 3×3 matrix with $\det(A) = 5$, what is $\det(2A)$?

a. 40 ✓

b. 10

c. 20

d. 80

For an $n \times n$ matrix, $\det(cA) = c^n \det(A)$. Here $n = 3$, so $\det(2A) = 2^3 \times 5 = 8 \times 5 = 40$.

© ClipSAT — clipsat.org | All Rights Reserved

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the eigenvalues of the matrix $A = [[4, 2], [1, 3]]$.

a. $\lambda = 5$ **and** $\lambda = 2$ ✓

b. $\lambda = 4$ and $\lambda = 3$

c. $\lambda = 6$ and $\lambda = 1$

d. $\lambda = 7$ and $\lambda = 0$

Characteristic equation: $\det(A - \lambda I) = (4 - \lambda)(3 - \lambda) - 2 = \lambda^2 - 7\lambda + 12 - 2 = \lambda^2 - 7\lambda + 10 = 0 \Rightarrow (\lambda - 5)(\lambda - 2) = 0$. Roots are $\lambda = 5, 2$.

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find an eigenvector corresponding to $\lambda = 5$ for $A = [[4, 2], [1, 3]]$.

a. $[2, 1]^T$ ✓

b. $[1, 2]^T$

c. $[1, -1]^T$

d. $[2, -1]^T$

Solve $(A - 5I)\mathbf{v} = \mathbf{0} \Rightarrow [[-1, 2], [1, -2]][x, y]^T = [0, 0]^T \Rightarrow -x + 2y = 0 \Rightarrow x = 2y$. With $y = 1$, $\mathbf{v} = [2, 1]^T$.

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the sum of the eigenvalues of an $n \times n$ matrix A equal to?

a. **The trace of A (sum of diagonal entries)** ✓

b. The determinant of A

c. The rank of A

d. n

By algebraic properties of the characteristic polynomial, $\sum \lambda_i = \text{tr}(A)$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

© ClipSAT — clipsat.org | All Rights Reserved

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

If a matrix A is diagonalized as $A = PDP^{-1}$, how is the power A^n computed efficiently?

a. $A^n = PD^nP^{-1}$ ✓

b. $A^n = P^nD^n(P^{-1})^n$

c. $A^n = D^n$

d. $A^n = PDP^{-1}n$

$A^n = (PDP^{-1})(PDP^{-1}) \ldots = PD^nP^{-1}$. For a diagonal matrix D, D^n simply raises each diagonal entry to the nth power.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

When is an $n \times n$ matrix A guaranteed to be diagonalizable?

a. **When it has n linearly independent eigenvectors (or n distinct eigenvalues)** ✓

b. When its determinant is non-zero

c. Only if it is a symmetric identity matrix

d. When all its entries are positive

A matrix is diagonalizable if and only if there exists an eigenbasis consisting of n linearly independent eigenvectors.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Consider the region R in the xy-plane bounded by the curves $y = x^2$ and $y = 4$. What is the area of this region?

a. $\frac{32}{3}$ ✓

b. $\frac{16}{3}$

c. 8

d. 16

The area of the region bounded by two curves $y = f(x)$ and $y = g(x)$ from $x = a$ to $x = b$, where $f(x) \geq g(x)$ on $[a, b]$, is given by the integral $\int_a^b (f(x) - g(x))dx$. First, we find the intersection points of $y = x^2$ and $y = 4$ by setting them equal: $x^2 = 4$, which gives $x = -2$ and $x = 2$. So, our limits of integration are $a = -2$ and $b = 2$. On the interval $[-2, 2]$, the line $y = 4$ is above the parabola $y = x^2$. Therefore, the area A is given by: $A = \int_{-2}^{2} (4 - x^2)dx$ Now, we eva.

© ClipSAT — clipsat.org | All Rights Reserved

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Consider the solid E bounded by the surfaces $z = x^2 + y^2$ and $z = 8 - x^2 - y^2$. Calculate the volume of this solid using a triple integral by first transforming to cylindrical coordinates. Express your answer in terms of π.

a. $\frac{16\pi}{3}$

b. $\frac{32\pi}{3}$ ✓

c. 16π

d. $\frac{64\pi}{3}$

The solid E is bounded below by the paraboloid $z = x^2 + y^2$ and above by the paraboloid $z = 8 - x^2 - y^2$. To find the intersection of these two surfaces, we set them equal: $x^2 + y^2 = 8 - x^2 - y^2$, which simplifies to $2(x^2 + y^2) = 8$, or $x^2 + y^2 = 4$. This is a circle of radius 2 in the xy-plane. In cylindrical coordinates, $x = r\cos(\theta)$, $y = r\sin(\theta)$, and $z = z$. The equations of the surfaces become $z = r^2$ and $z = 8 - r^2$. The intersection occurs at $r^2 = 8 - r^2$, so.

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Two lines have slopes $m_1 = -1$, $m_2 = -1$ and y-intercepts $b_1 = -2$, $b_2 = -2$. Classify the system.

a. Consistent and independent (one solution)

b. **Consistent and dependent (infinitely many solutions)** ✓

c. Inconsistent (no solution)

d. Cannot be determined

Different slopes give exactly one intersection (independent). Equal slopes with equal intercepts give the same line (dependent, infinite solutions). Equal slopes with different intercepts give parallel lines (inconsistent, no solution).

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Probability: $f(x, y) = 6xy$ on $0 \le x \le 1, 0 \le y \le \sqrt{1 - x^2}$ (quarter circle). $P(X + Y \le 1) =$

a. $\frac{3}{4}$

b. $\frac{1}{2}$

c. $\frac{1}{4}$ ✓

d. $\frac{2}{3}$

Integrating over the triangle $x \ge 0, y \ge 0, x + y \le 1$ gives $\int_0^1 \int_0^{1-x} 6xy\, dy\, dx = 3\int_0^1 x(1-x)^2\, dx = 3\left(\frac{1}{2} - \frac{2}{3} + \frac{1}{4}\right) = \frac{1}{4}$.

© ClipSAT — clipsat.org | All Rights Reserved

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Evaluate $\int_0^2 \int_y^2 e^{x^2}\,dx\,dy$ by reversing the order of integration.

a. $-\frac{1}{2} + \frac{e^4}{2}$ ✓

b. $-1 + e^4$

c. $\frac{1}{2} + \frac{e^4}{2}$

d. $-\frac{1}{4} + \frac{e^4}{4}$

The inner integral has no elementary antiderivative as written. Reversing the order gives $\int_0^2 \int_0^x e^{x^2}\,dy\,dx$, which evaluates to $-\frac{1}{2} + \frac{e^4}{2}$.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Use $x = u + v, y = u - v$ to compute $\iint_D (x - y)e^{x+y}\,dA$ where D is the square $|x - y| \le 1, |x + y| \le 1$:

a. 0 ✓

b. $2(e - e^{-1})$

c. $(e - 1)^2$

d. $4(e - e^{-1})/e$

Under the change of variables $2u = x + y$ and $2v = x - y$, the integrand $2v\,e^{2u}$ is odd in v over the symmetric interval $[-1/2, 1/2]$, so the double integral equals 0.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Iterated integral over tetrahedron $x + y + z \le 1, x, y, z \ge 0$: volume = $1/6$. The integral $\iiint x\,dV =$

a. $1/6$

b. $1/24$ ✓

c. $1/12$

d. $1/8$

By symmetry $\iiint x = \iiint y = \iiint z$; and $\iiint (x + y + z) = 3\iiint x$; by integration $\iiint (x + y + z)dV = 1/8$... actually $\iiint x\,dV = 1/24$ (standard result).

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)

Evaluate $\iiint_B (x^2 + y^2 + z^2)\,dV$ where B is the ball $x^2 + y^2 + z^2 \le 1$.

a. $\frac{4\pi}{5}$ ✓

b. $\frac{4\pi}{3}$

c. $\frac{2\pi}{5}$

d. $\frac{8\pi}{5}$

In spherical coordinates the integrand is ρ^2, so $\int_0^{2\pi} \int_0^{\pi} \int_0^1 \rho^4 \sin\phi\,d\rho\,d\phi\,d\theta = \frac{4\pi}{5}$.

© ClipSAT — clipsat.org | All Rights Reserved

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.03 Markov Chains (AI HL)

The volume of the region $1 \le x^2 + y^2 + z^2 \le 4$ (shell between spheres of radii 1 and 2):

a. $4\pi/3$

b. 4π

c. 8π

d. $4\pi(8-1)/3 = 28\pi/3$ ✓

Applying Multiple Integrals principles, $V = (4\pi/3)(2^3 - 1^3) = 28\pi/3$. Therefore, the verified correct option is $4\pi(8-1)/3 = 28\pi/3$.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.01 Proof by Mathematical Induction

Use spherical coordinates to find the volume of the first-octant part of the ball $x^2 + y^2 + z^2 \le 16$.

a. $\frac{256\pi}{3}$

b. $\frac{64\pi}{3}$

c. $\frac{32\pi}{3}$ ✓

d. $\frac{16\pi}{3}$

$dV = \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta$; integrating over the region gives $\frac{1}{8} \cdot \frac{4}{3}\pi(4)^3 = \frac{32\pi}{3}$.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra

Find the volume of the solid inside the sphere $\rho = 2$ and above the cone $\phi = \frac{\pi}{4}$.

a. $2\pi\left(\frac{8}{3} - \frac{4\sqrt{2}}{3}\right)$ ✓

b. $4\pi\left(\frac{8}{3} - \frac{4\sqrt{2}}{3}\right)$

c. $\frac{16\pi}{3}$

d. $2\sqrt{2}\pi\left(\frac{8}{3} - \frac{4\sqrt{2}}{3}\right)$

$V = \int_0^{2\pi}\int_0^{\pi/4}\int_0^{2} \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta = 2\pi\left(\frac{8}{3} - \frac{4\sqrt{2}}{3}\right)$.

© ClipSAT — clipsat.org | All Rights Reserved

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity

The Jacobian of the transformation $x = r\cosh u, y = r\sinh u$ is r times:

a. $\cosh^2 u - \sinh^2 u = 1$ ✓

b. $\cosh u + \sinh u$

c. $\sinh^2 u$

d. r^2

Applying Multiple Integrals principles, $J = r(\cosh^2 u - \sinh^2 u) = r$. Therefore, the verified correct option is $\cosh^2 u - \sinh^2 u = 1$.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.04 Partial Fractions

Centre of mass of lamina $D = [0,1] \times [0,1]$ with density $\delta = xy$:

a. $(1/2, 1/2)$

b. $(2/3, 2/3)$ ✓

c. $(3/4, 3/4)$

d. $(1, 1)$

$m = \int_0^1 \int_0^1 xy\,dy\,dx = 1/4$; $M_y = \int_0^1 \int_0^1 x^2 y\,dy\,dx = 1/6$; $\bar{x} = (1/6)/(1/4) = 2/3$; similarly $\bar{y} = 2/3$.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)

Use polar coordinates to evaluate $\iint_R (x^2 + y^2)^3\,dA$, where R is the disk $x^2 + y^2 \le 9$.

a. $\frac{6561}{4}$

b. $\frac{6561\pi}{4}$ ✓

c. $\frac{6561\pi}{2}$

d. $\frac{6561\pi}{8}$

With $x = r\cos\theta,\ y = r\sin\theta$ and $dA = r\,dr\,d\theta$, the integral becomes elementary; its value is $\frac{6561\pi}{4}$.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem

The double integral over region D of $f = 1$ gives:

a. **area of** D ✓

b. perimeter of D

c. volume

d. 0

Applying Multiple Integrals principles, $\iint_D 1\,dA = \text{Area}(D)$. Therefore, the verified correct option is area of D.

© ClipSAT — clipsat.org | All Rights Reserved